

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

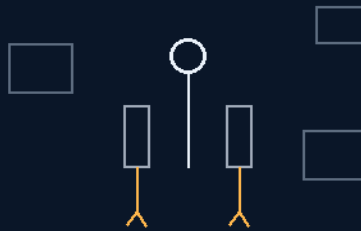
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 185

QUINN'S THE IRON MAN CANISTER COURSE

Hand-held pressure cans as thrusters
around, up and over in 3D



no combustion in the crew's air

COLD GAS · REFILLABLE · SHIP'S AIR
FLYING ON WHAT THE SHIP BREATHES

Recreation system — v1.0 in force

GP-IM-185

Revision v1.0 — 23 August 2026

186 of 290

VETTED BATCH 22 — PASS · BUILD SHEET OWED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 185

PHASE 185: QUINN'S THE IRON MAN CANISTER COURSE — COLD-GAS FLIGHT CLUB — STATUS: CURRENT — PASS; THRUST DURATION, GAS CONSUMPTION, NOZZLE TORQUE AND PRESSURE-VESSEL SIZING ARE BUILD-SHEET NUMBERS. The purest zero-G sport of the six: take away gravity and the obstacle course goes three-dimensional — around, up and over costs only momentum, not altitude. Phase 185 is flight club: hand-held refillable pressure cans as personal thrusters. Cold gas wins the room — no combustion, no propellant chemistry, no fire risk in an atmosphere the crew breathes. Refillable by doctrine: the cans recharge from the ship's own compressed-air infrastructure — no consumable cartridges, no supply chain; the rec deck runs on the ship's air the way the galley runs on its water. The vet passed it, and named the build-sheet numbers: thrust duration, gas consumption, nozzle torque and pressure-vessel sizing — the owed work, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-185, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the flight law as seated (contents line, the body — the thruster — the audit and provenance, verbatim) — §2 why cold gas wins the room (graded) — §3 the system in the can — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 185: **The Iron Man Canister Course — Cold-Gas Flight Club.** Hand-held refillable pressure cans as personal thrusters around, up and over a three-dimensional zero-G obstacle course — cold gas only (no combustion, hull-safe), refilled from the ship's own compressed-air infrastructure. ~10N handheld thrust moves a 90kg rider at 0.11 m/s^2 — zippy in zero-G, nowhere near Earth hover, and the file says so. Torque training, insulated grips, ship-rated pressure vessels flagged.*'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Human Flight Around, Up and Over an Obstacle Course — hand-held thrusters, no tethers, no combustion '

Crew Life Baseline: Personal Cold-Gas Maneuvering Sport (Zero-G)

Live instruction, 30 July 2026. Fifth game of the rec-deck set. The purest zero-G sport of the six: take away gravity and the only thing between a crew member and flight is a thrust source safe enough to hold in your hand. His answer: refillable pressure cans.

THE THRUSTER — WHY COLD GAS WINS THE ROOM (VERBATIM)

- **Hand-held refillable pressure cans:** each rider carries pressure cans as thrusters — cold gas, no combustion, no propellant chemistry, no fire risk in a pressure hull. Squeeze the valve, the gas leaves, you move. Newton's third law, handheld.
- **Refillable by doctrine:** the cans recharge from the ship's own compressed-air infrastructure — no consumable cartridges, no supply chain. The rec deck's compressor station is a shared utility: one fill point serves this course and Phase 186's rollerball cans alike.
- **The course:** around, up and over an obstacle course — in zero-G 'up and over' costs only momentum, not altitude, so the course is a three-dimensional line instead of a floor plan. Rings, gates, weaves and over-the-top passes; the layout is movable rec-deck hardware.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED — COLD-GAS MANOEUVRING IS FLOWN SPACE HARDWARE, AND THE NUMBERS FIT A HUMAN HAND) (VERBATIM)

Cold-gas human manoeuvring is not theory: astronaut rescue and manoeuvring units have flown pressurised-gas jets for decades, because a gas jet needs no ignition, no fuel handling and no exhaust hazard beyond the gas itself — which is why it is the only thrust source sane people hand to a human indoors. The arithmetic fits the sport: thrust is mass-flow times exhaust velocity, and even a modest 10 newtons from a hand-held can accelerates a 90 kg rider at about 0.11 m/s^2 — in zero-G that is genuinely zippy, building a metre per second in nine seconds, which is all a tight course can use. The same 10 N is the honest boundary of the game: against Earth's gravity a human needs roughly 900 N just to hover, so this is a zero-G sport — on Earth the cans are ground-assist at best, and the file records that boundary rather than blurring it. Flags, kept honest: (1) a hand-held thruster fires off the body's centre of mass, so every squeeze is also a torque — untrained riders will spin; technique and two-hand balancing are training matters, and Phase 186's two-can doctrine is the natural trainer. (2) Sustained discharge chills the can by adiabatic expansion — insulated grips are a build spec, not an accessory. (3) Cans are pressure vessels: ship-rated cylinders and burst protection are engineering specs for the build sheet, flagged here so they are never treated as toys. None of the flags ground the game.

ORIGIN OF THOUGHT — PHASE 185 (VERBATIM)

Archer, 30 July 2026: "iron man obstacle course, hand held refillable pressure cans as thrusters around up and over an obstacle course"

Why it matters, in plain English: this is the game that teaches the crew to fly. Every other sport on the deck plays with the air; this one plays with Newton. A crew that has spent a year dodging gates on hand-held thrusters is a crew that moves well in zero-G — which is a safety outcome wearing a sports uniform. And note the name: he called it Iron Man, and then built it from the one propulsion system that actually works indoors. The movies get the suit; the ship gets the physics.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

COLD-GAS MANOEUVRING IS FLOWN SPACE HARDWARE (the audit's affirmation): astronaut manoeuvring units flew on compressed nitrogen — hand-held thrust in a pressurized volume is not exotic, it is heritage. The canister course is a flown technology retired into a sport.

THE ATMOSPHERE ARGUMENT, AFFIRMED: inside the hull, any combustion or monopropellant spends the crew's air and adds chemistry to it. Cold gas — the ship's own compressed air — adds nothing and costs nothing the ship was not already carrying.

THE 3D COURSE, AFFIRMED AS GAME DESIGN: in free fall, 'up and over' is a momentum problem, not an altitude problem — the obstacle course gains its third dimension for free, and flight lines replace climbing holds.

THE NOZZLE-TORQUE FLAG, AFFIRMED AS REAL: a hand-held thruster off the body's centreline torques the wrist and spins the rider — the vet's build-sheet numbers (duration, consumption, torque, vessel sizing) are exactly the right owed list, seated in §9.

§3 — THE SYSTEM IN THE CAN

- THE CANS: hand-held refillable pressure cans — cold gas, personal thrusters.
- THE FILL: the ship's compressed-air infrastructure — no cartridges, no supply chain.
- THE COURSE: around, up and over — three-dimensional in zero-G.
- THE CLUB: flight as sport — the rec deck's purest zero-G game.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE COLD-GAS DOCTRINE: no combustion in the crew's air — the thruster breathes what the ship breathes.

- THE REFILLABLE DOCTRINE: the cans recharge from ship's air — the rec deck carries no consumables.
- THE HERITAGE DOCTRINE: flown hardware retired into play — the manoeuvring unit's second career.
- THE MOMENTUM-NOT-ALTITUDE DOCTRINE: zero-G courses are 3D for free.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Spec the can: pressure-vessel sizing and rating to the fleet's air standard (the vet's numbers).
- STEP 2 — Bench the thrust: duration and gas consumption per fill; nozzle torque at the wrist mapped.
- STEP 3 — Build the course: obstacles, flight lines, safety netting.
- STEP 4 — Flight trials: rider control, refill cadence, course times.
- STEP 5 — Publish the can spec and the course — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 186 — curved-track rollerball (GP-IM-186): the same cans, the same fill station, on rails.
- PHASE 183 — battlestar (GP-IM-183): the rec-deck set's court game.
- PHASE 173 — the battery mandate (GP-IM-173): the compressors ride fleet power.
- PHASE 165 — the tether gauntlets (GP-IM-165): the line-work cousin of free flight.

§8 — DO-NOT-BUILD

- DO NOT combust — cold gas only; the crew breathes the course.
- DO NOT cartridge — refillable from ship's air; no consumable supply chain.
- DO NOT ignore the torque — off-centreline thrust spins the rider; the build sheet is law.
- DO NOT unrated vessels — pressure sizing to the fleet standard, every can.

§9 — OWED

OWED (the vet's yellow): the build-sheet numbers — thrust duration, gas consumption, nozzle torque, pressure-vessel sizing (§6).

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-185, v1.0 — CURRENT — PASS; the build-sheet numbers owed, seated law, master v2.1 (numerical print order). Built 23 August 2026, eighteenth of the 20-chapter 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 185